

Quadratics

Student Question Booklet

Chapters 5 & 7 : Year 10 Mathematical Methods : 2026

How to use this booklet

This booklet covers the whole quadratic story in two parts:

- **Part A — Quadratic Algebra (Chapter 5).** Expanding, factorising, and solving quadratic equations.
- **Part B — Quadratic Functions (Chapter 7).** Parabolas, sketching, applications, rates of change and variation.

For every section you will find:

- **Worked examples** — fully solved, with a diagram where it helps, plus the matching textbook example numbers and pages.
- **Question sets** — what to practise, grouped by *Building Understanding*, *Fluency*, *Problem-solving*, *Reasoning* and *Enrichment*, with page numbers. Find them in your own textbook.
- **Hurdle questions** — five questions you **must** be able to do to pass the section, with working space.

Diagrams are schematic (not to scale). Sections marked 10A are extension. Always show full working — in algebra the method earns the marks.

Page references are to *Essential Mathematics for the Victorian Curriculum 10 & 10A*, 3rd ed. Question sets belong to the textbook; the worked examples, diagrams and hurdle questions are original material written for this course.

Part A — Quadratic Algebra

Chapter 5 : expanding, factorising and solving quadratic equations

5A Expanding expressions

CONSOLIDATING

Textbook worked examples (cross-reference in your textbook)

- **Example 1** — Expanding simple expressions (p. 417)
- **Example 2** — Expanding binomial products, perfect squares and DOPS (p. 418)
- **Example 3** — Expanding more binomial products (p. 418)

Question sets (practise from your textbook)

- **Building Understanding:** Q1–3 (p. 417)
- **Fluency:** Q1–5 (p. 419–420)
- **Problem-solving:** Q6–9 (p. 420)
- **Reasoning:** Q10–13 (p. 421)
- **Enrichment: Expanding to prove:** Q14 (p. 421)

Worked examples (study the method, then try the hurdles)

Expanding a binomial product

Expand and simplify $(x + 5)(x - 2)$.

	x	$+5$
x		
-2		

Solution. Multiply every term in the first bracket by every term in the second (FOIL), then collect like terms.

$$\begin{aligned}(x + 5)(x - 2) &= x^2 - 2x + 5x - 10 \\ &= x^2 + 3x - 10.\end{aligned}$$

Perfect square and difference of squares

Expand **(a)** $(x + 4)^2$ and **(b)** $(x - 3)(x + 3)$.

Solution. **(a)** $(a + b)^2 = a^2 + 2ab + b^2$:

$$(x + 4)^2 = x^2 + 8x + 16.$$

(b) $(a - b)(a + b) = a^2 - b^2$, so the middle terms cancel:

$$(x - 3)(x + 3) = x^2 - 9.$$

Hurdle questions — Section 5A (show all working)

H1. Expand and simplify:

(a) $3(2x - 5)$

	$2x$	-5
3		

(b) $-2x(4x + 7)$

	$4x$	$+7$
$-2x$		

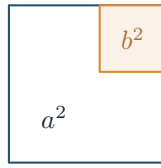
H2. Expand and simplify $(x + 4)(x - 3)$.

	x	$+4$
x		
-3		

H3. Expand the perfect square $(2x + 5)^2$.

	$2x$	$+5$
$2x$		
$+5$		

H4. Expand the difference of perfect squares $(3x - 2)(3x + 2)$.



$$a^2 - b^2 = (a - b)(a + b)$$

H5. Expand and simplify $(x + 3)(x - 2) - (x - 1)(x + 4)$.

5B Factorising expressions

CORE

Textbook worked examples (cross-reference in your textbook)

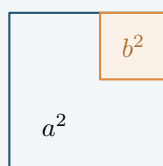
- **Example 4** — Taking out common factors (p. 423)
- **Example 5** — Factorising a difference of two squares (p. 424)
- **Example 6** — Factorising a DOPS using surds (p. 424)
- **Example 7** — Factorising by grouping (p. 425)

Question sets (practise from your textbook)

- **Building Understanding:** Q1–3 (p. 423)
- **Fluency:** Q1–6 (p. 425–426)
- **Problem-solving:** Q7–10 (p. 426)
- **Reasoning:** Q11–14 (p. 427)
- **Enrichment: Hidden difference of two squares:** Q15–16 (p. 427)

Worked examples (study the method, then try the hurdles)**Taking out a common factor**Factorise $12x^2 - 18x$.**Solution.** The highest common factor of $12x^2$ and $18x$ is $6x$. Divide each term by it:

$$12x^2 - 18x = 6x(2x - 3).$$

Difference of two squaresFactorise $4x^2 - 25$.

$$a^2 - b^2 = (a - b)(a + b)$$

Solution. Write each term as a square: $4x^2 = (2x)^2$ and $25 = 5^2$. Then use $a^2 - b^2 = (a - b)(a + b)$:

$$4x^2 - 25 = (2x - 5)(2x + 5).$$

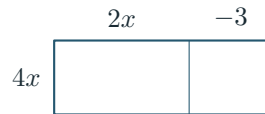
Hurdle questions — Section 5B (show all working)

H1. Factorise by taking out the highest common factor:

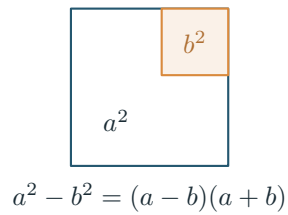
(a) $6x + 18$



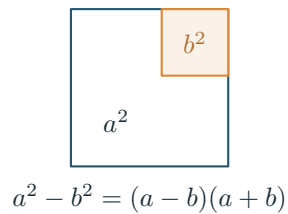
(b) $8x^2 - 12x$



H2. Factorise the difference of two squares $x^2 - 49$.



H3. Factorise $9x^2 - 25$.



H4. Factorise $x^2 - 7$ using surds.

H5. Factorise by grouping: $x^2 + 3x + 2x + 6$.

5C Multiplying and dividing algebraic fractions

CORE

Textbook worked examples (cross-reference in your textbook)

- **Example 8** — Cancelling common factors (p. 429)
- **Example 9** — Multiplying and dividing algebraic fractions (p. 430)

Question sets (practise from your textbook)

- **Building Understanding:** Q1–3 (p. 429)
- **Fluency:** Q1–4 (p. 430–431)
- **Problem-solving:** Q5–8 (p. 431)
- **Reasoning:** Q9–11 (p. 432)
- **Enrichment: Same or different?:** Q12 (p. 432)

Worked examples (study the method, then try the hurdles)

Simplifying by cancelling

Simplify $\frac{x^2 - 4}{x - 2}$.

Solution. Factorise the numerator as a difference of two squares, then cancel:

$$\frac{x^2 - 4}{x - 2} = \frac{(x - 2)(x + 2)}{x - 2} = x + 2, \quad x \neq 2.$$

Multiplying algebraic fractions

Simplify $\frac{3x}{4} \times \frac{8}{9x^2}$.

Solution. Cancel common factors before multiplying:

$$\frac{3x}{4} \times \frac{8}{9x^2} = \frac{3x \times 8}{4 \times 9x^2} = \frac{24x}{36x^2} = \frac{2}{3x}.$$

Hurdle questions — Section 5C (show all working)

H1. Simplify by cancelling common factors: $\frac{6x^2}{9x}$.

H2. Simplify $\frac{3(x+2)}{6(x+2)}$.

H3. Simplify $\frac{x^2-9}{x+3}$.

H4. Simplify $\frac{2x}{5} \times \frac{15}{4x^2}$.

H5. Simplify $\frac{x+1}{4} \div \frac{x+1}{8}$.

5D Factorising monic quadratic trinomials

CORE

Textbook worked examples (cross-reference in your textbook)

- **Example 10** — Factorising trinomials of the form $x^2 + bx + c$ (p. 434)
- **Example 11** — Simplifying algebraic fractions (p. 435)

Question sets (practise from your textbook)

- **Building Understanding:** Q1–3 (p. 434)
- **Fluency:** Q1–4 (p. 435–436)
- **Problem-solving:** Q5–7 (p. 436)
- **Reasoning:** Q8–11 (p. 436)
- **Enrichment: Addition and subtraction with factorisation:** Q12 (p. 437)

Worked examples (study the method, then try the hurdles)

Factorising a monic trinomial

Factorise $x^2 + 9x + 20$.

	x	$+4$
x		
$+5$		

Solution. Find two numbers that *multiply* to 20 and *add* to 9: these are 4 and 5.

$$x^2 + 9x + 20 = (x + 4)(x + 5).$$

A trinomial with a negative constant

Factorise $x^2 - 3x - 10$.

Solution. Two numbers multiplying to -10 and adding to -3 : these are -5 and 2 .

$$x^2 - 3x - 10 = (x - 5)(x + 2).$$

Hurdle questions — Section 5D (show all working)

H1. Factorise $x^2 + 7x + 12$.

	x	$+3$
x		
$+4$		

H2. Factorise $x^2 - 5x + 6$.

H3. Factorise $x^2 + 2x - 15$.

H4. Factorise $x^2 - 9x - 22$.

H5. Simplify $\frac{x^2 + 5x + 6}{x + 2}$ by factorising first.

5E Factorising non-monic quadratic trinomials

10A

Textbook worked examples (cross-reference in your textbook)

- **Example 12** — Factorising non-monic quadratics (p. 439)
- **Example 13** — Simplifying algebraic fractions involving quadratics (p. 440)

Question sets (practise from your textbook)

- **Building Understanding:** Q1–3 (p. 439)
- **Fluency:** Q1–3 (p. 440)
- **Problem-solving:** Q4–6 (p. 440)
- **Reasoning:** Q7–9 (p. 441)
- **Enrichment: Non-monics with addition and subtraction:** Q10 (p. 441)

Worked examples (study the method, then try the hurdles)**Factorising a non-monic trinomial**Factorise $3x^2 + 11x + 6$.**Solution.** Multiply $a \times c = 3 \times 6 = 18$; find two numbers multiplying to 18 and adding to 11: 9 and 2. Split the middle term and group:

$$\begin{aligned}
 3x^2 + 11x + 6 &= 3x^2 + 9x + 2x + 6 \\
 &= 3x(x + 3) + 2(x + 3) \\
 &= (x + 3)(3x + 2).
 \end{aligned}$$

Hurdle questions — Section 5E (show all working)**H1.** Factorise $2x^2 + 7x + 3$.

H2. Factorise $3x^2 - 10x + 8$.

H3. Factorise $6x^2 + x - 2$.

H4. Factorise $4x^2 - 8x - 5$.

H5. Simplify $\frac{2x^2 + 5x + 2}{2x + 1}$ by factorising first.

5F Factorising by completing the square

CORE

Textbook worked examples (cross-reference in your textbook)

- **Example 14** — Completing the square (p. 444)
- **Example 15** — Factorising by completing the square (p. 445)
- **Example 16** — Factorising with fractions and non-monics (p. 445)

Question sets (practise from your textbook)

- **Building Understanding:** Q1–3 (p. 444)
- **Fluency:** Q1–4 (p. 446)
- **Problem-solving:** Q5–7 (p. 447)
- **Reasoning:** Q8–9 (p. 447)
- **Enrichment: Proof by completing the square:** Q10 (p. 447)

Worked examples (study the method, then try the hurdles)**Completing the square**Write $x^2 + 8x$ in the form $(x + a)^2 - b$.

x^2	$\frac{b}{2}x$
$\frac{b}{2}x$	$(\frac{b}{2})^2$

Solution. Halve the coefficient of x and square it: $(\frac{8}{2})^2 = 16$. Add and subtract it:

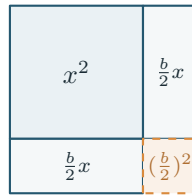
$$x^2 + 8x = (x + 4)^2 - 16.$$

Factorising by completing the squareFactorise $x^2 + 6x + 4$ by completing the square.**Solution.** Half of 6 is 3, and $3^2 = 9$:

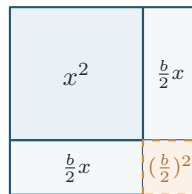
$$\begin{aligned} x^2 + 6x + 4 &= (x + 3)^2 - 9 + 4 = (x + 3)^2 - 5 \\ &= (x + 3 - \sqrt{5})(x + 3 + \sqrt{5}). \end{aligned}$$

Hurdle questions — Section 5F (show all working)

H1. Complete the square: write $x^2 + 6x$ in the form $(x + a)^2 - b$.



H2. Complete the square: write $x^2 - 8x$ in the form $(x + a)^2 - b$.



H3. Factorise $x^2 + 4x - 1$ by completing the square (surds in the answer).

H4. Factorise $x^2 - 6x + 2$ by completing the square.

H5. Write $x^2 + 5x$ in completed-square form (fractions required).

5G Solving quadratic equations using factorisation

CORE

Textbook worked examples (cross-reference in your textbook)

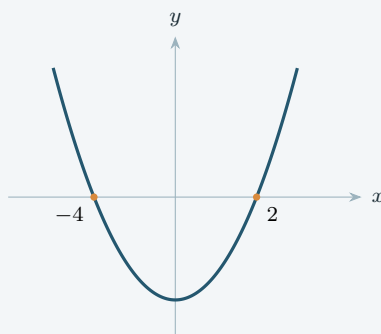
- **Example 17** — Solving using the Null Factor Law (p. 449)
- **Example 18** — Solving $ax^2 + bx + c = 0$ (p. 450)
- **Example 19** — Solving disguised quadratics (p. 451)

Question sets (practise from your textbook)

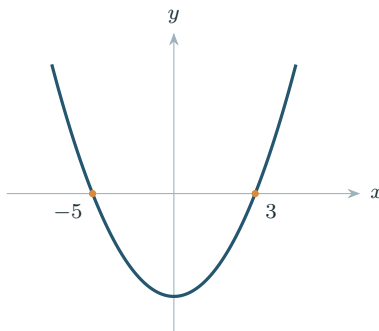
- **Building Understanding:** Q1–3 (p. 449)
- **Fluency:** Q1–4 (p. 452)
- **Problem-solving:** Q5–8 (p. 452)
- **Reasoning:** Q9–11 (p. 453)
- **Enrichment: More quadratics in disguise:** Q12 (p. 453)

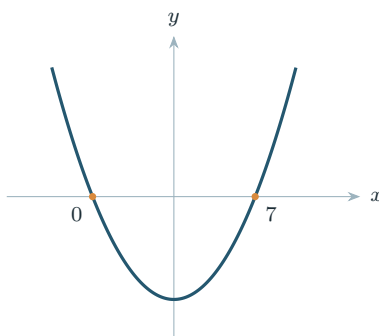
Worked examples (study the method, then try the hurdles)**Using the Null Factor Law**Solve $(x - 4)(x + 7) = 0$.**Solution.** If a product is zero, at least one factor is zero:

$$x - 4 = 0 \text{ or } x + 7 = 0 \Rightarrow x = 4 \text{ or } x = -7.$$

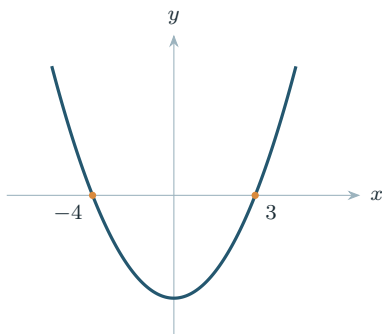
Solving by factorising firstSolve $x^2 + 2x - 8 = 0$.**Solution.** Factorise, then apply the Null Factor Law:

$$x^2 + 2x - 8 = (x + 4)(x - 2) = 0 \Rightarrow x = -4 \text{ or } x = 2.$$

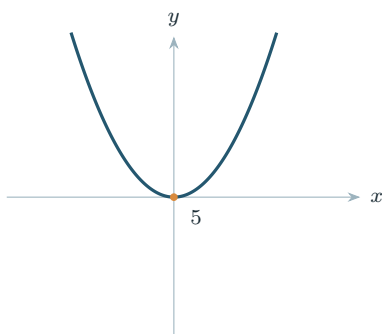
Hurdle questions — Section 5G (show all working)**H1.** Solve $(x - 3)(x + 5) = 0$ using the Null Factor Law.

H2. Solve $x^2 - 7x = 0$ by factorising.

H3. Solve $x^2 + x - 12 = 0$.



H4. Solve $x^2 - 10x + 25 = 0$.



H5. Solve $2x^2 + 5x + 3 = 0$.

5H Applications of quadratic equations

CORE

Textbook worked examples (cross-reference in your textbook)

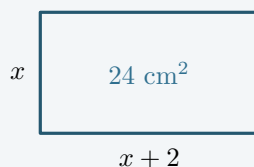
- **Example 20** — Finding dimensions using a quadratic equation (p. 455)

Question sets (practise from your textbook)

- **Building Understanding:** Q1–2 (p. 455)
- **Fluency:** Q1–6 (p. 456)
- **Problem-solving:** Q7–11 (p. 456)
- **Reasoning:** Q12–14 (p. 457)
- **Enrichment: Fixed perimeter and area:** Q15–16 (p. 457)

Worked examples (study the method, then try the hurdles)**Finding dimensions with a quadratic**

A rectangle is 2 cm longer than it is wide and has area 24 cm^2 . Find its dimensions.



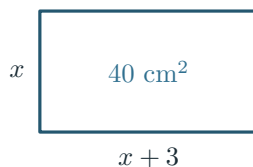
Solution. Let the width be x , so the length is $x + 2$. Then

$$\begin{aligned} x(x + 2) &= 24 \\ x^2 + 2x - 24 &= 0 \\ (x + 6)(x - 4) &= 0. \end{aligned}$$

So $x = 4$ (reject $x = -6$, a length cannot be negative). The rectangle is 4 cm by 6 cm.

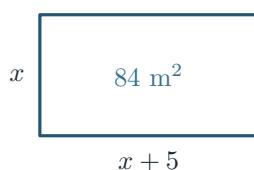
Hurdle questions — Section 5H (show all working)

- H1.** A rectangle is 3 cm longer than it is wide and has an area of 40 cm^2 . Letting the width be x , write an equation and solve it to find the dimensions.

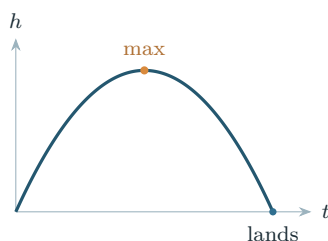


- H2.** The product of two consecutive positive integers is 56. Set up a quadratic equation and find the integers.

- H3.** A rectangular garden has a length 5 m more than its width and an area of 84 m^2 . Find its dimensions.



- H4.** A ball is thrown so its height is $h = 20t - 5t^2$ metres after t seconds. Find the times when the ball is at ground level ($h = 0$).



- H5.** The square of a number, less four times the number, equals 21. Find all possible values of the number.

5I Solving quadratic equations by completing the square

CORE

Textbook worked examples (cross-reference in your textbook)

- **Example 21** — Solving quadratic equations by completing the square (p. 461)

Question sets (practise from your textbook)

- **Building Understanding:** Q1–2 (p. 461)
- **Fluency:** Q1–3 (p. 462)
- **Problem-solving:** Q4–8 (p. 463–464)
- **Reasoning:** Q9–11 (p. 464)
- **Enrichment: Completing the rectangle:** Q12 (p. 464)

Worked examples (study the method, then try the hurdles)**Solving by completing the square**Solve $x^2 + 4x - 3 = 0$, leaving your answer in surd form.**Solution.** Half of 4 is 2, and $2^2 = 4$:

$$\begin{aligned}
 x^2 + 4x - 3 &= 0 \\
 (x + 2)^2 - 4 - 3 &= 0 \\
 (x + 2)^2 &= 7 \\
 x + 2 &= \pm\sqrt{7} \\
 x &= -2 \pm \sqrt{7}.
 \end{aligned}$$

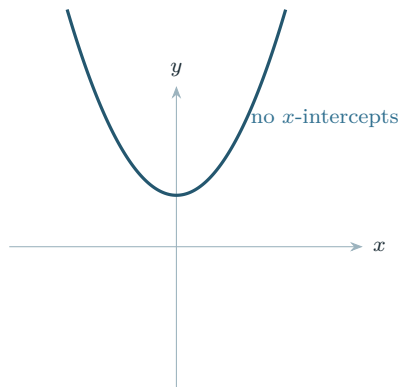
Hurdle questions — Section 5I (show all working)**H1.** Solve $x^2 + 6x + 5 = 0$ by completing the square.

H2. Solve $x^2 - 4x - 1 = 0$ by completing the square, leaving your answer in surd form.

H3. Solve $x^2 + 2x - 5 = 0$ by completing the square, leaving your answer in surd form.

H4. Solve $x^2 - 8x + 3 = 0$ by completing the square.

H5. Explain why $x^2 + 2x + 7 = 0$ has no real solutions, using completing the square.



5J Solving quadratic equations using the quadratic formula

CORE

Textbook worked examples (cross-reference in your textbook)

- **Example 22** — Using the discriminant (p. 467)
- **Example 23** — Solving using the quadratic formula (p. 468)

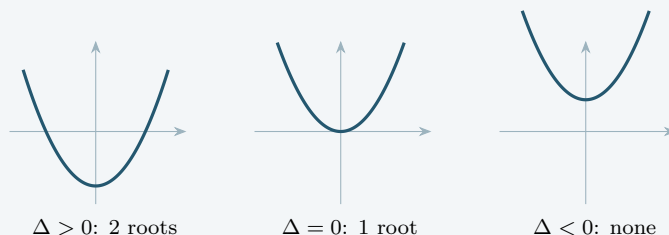
Question sets (practise from your textbook)

- **Building Understanding:** Q1–3 (p. 466)
- **Fluency:** Q1–3 (p. 469)
- **Problem-solving:** Q4–8 (p. 469)
- **Reasoning:** Q9–11 (p. 470)
- **Enrichment:** k determines the number of solutions: Q12 (p. 470)

Worked examples (study the method, then try the hurdles)

Using the discriminant

Find the discriminant of $x^2 - 4x + 7 = 0$ and state the number of real solutions.



Solution. With $a = 1$, $b = -4$, $c = 7$:

$$\Delta = b^2 - 4ac = (-4)^2 - 4(1)(7) = 16 - 28 = -12.$$

Since $\Delta < 0$, there are **no real solutions**.

Using the quadratic formula

Solve $2x^2 + 3x - 1 = 0$, correct to two decimal places.

Solution. With $a = 2$, $b = 3$, $c = -1$:

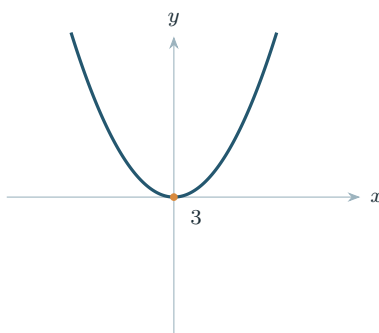
$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{-3 \pm \sqrt{9 + 8}}{4} = \frac{-3 \pm \sqrt{17}}{4}.$$

So $x = 0.28$ or $x = -1.78$ (2 dp).

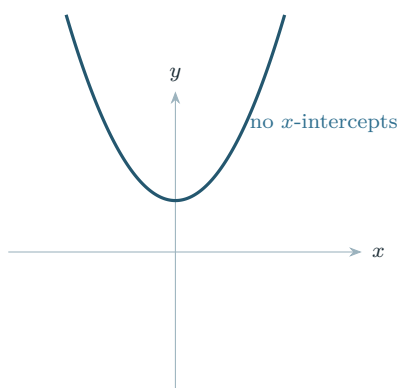
Hurdle questions — Section 5J (show all working)

H1. For $x^2 + 3x - 4 = 0$, state the values of a , b and c , then find the discriminant $\Delta = b^2 - 4ac$.

H2. Use the discriminant to state how many real solutions $x^2 - 6x + 9 = 0$ has.



H3. Use the discriminant to state how many real solutions $2x^2 + x + 5 = 0$ has.



H4. Solve $x^2 + 5x + 2 = 0$ using the quadratic formula, in exact (surd) form.

H5. Solve $3x^2 - 2x - 4 = 0$ using the quadratic formula, correct to two decimal places.

Part B — Quadratic Functions

Chapter 7 : parabolas, sketching, applications, rates of change and variation

7A Exploring parabolas

CORE

Textbook worked examples (cross-reference in your textbook)

- **Example 1** — Identifying key features of parabolas (p. 588)
- **Example 2** — Transforming parabolas (p. 589)

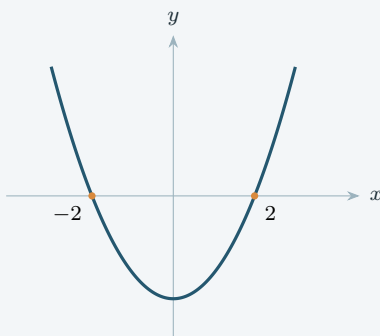
Question sets (practise from your textbook)

- **Building Understanding:** Q1–3 (p. 587)
- **Fluency:** Q1–4 (p. 590–592)
- **Problem-solving:** Q5–8 (p. 593)
- **Reasoning:** Q9–12 (p. 594)
- **Enrichment: Finding the rule:** Q13–14 (p. 594)

Worked examples (study the method, then try the hurdles)

Key features of a parabola

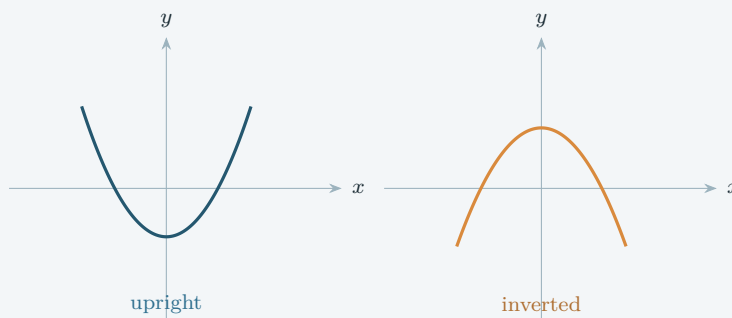
For $y = x^2 - 4$, state the turning point, the axis of symmetry, the y -intercept and the x -intercepts.



Solution. This is $y = x^2$ moved down 4 units. Turning point $(0, -4)$ (a minimum); axis of symmetry $x = 0$; y -intercept -4 . For x -intercepts set $y = 0$: $x^2 - 4 = 0$, so $x = \pm 2$.

Transforming parabolas

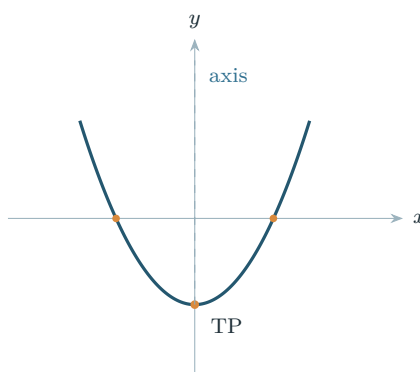
Describe how $y = -2x^2$ differs from $y = x^2$.



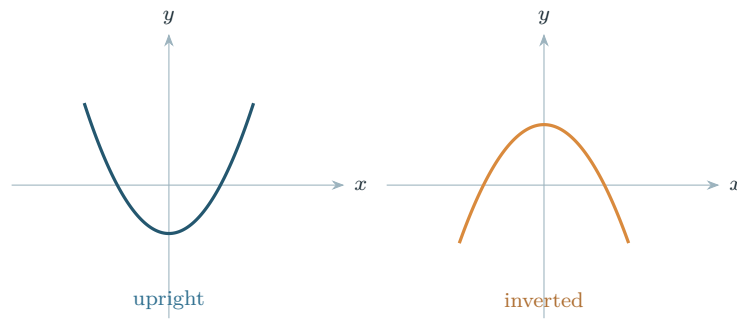
Solution. The negative sign *reflects* the parabola in the x -axis, so it is inverted (turning point is a maximum). The factor 2 makes it *narrower* than $y = x^2$, since each y -value is doubled.

Hurdle questions — Section 7A (show all working)

- H1.** For the parabola shown, state the coordinates of the turning point, the axis of symmetry, and the y -intercept.



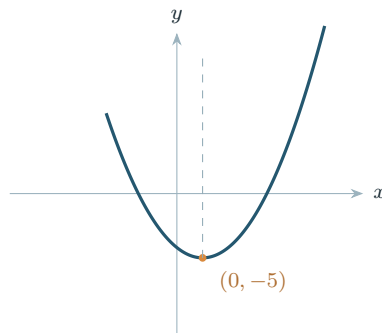
H2. State whether each parabola is upright or inverted:



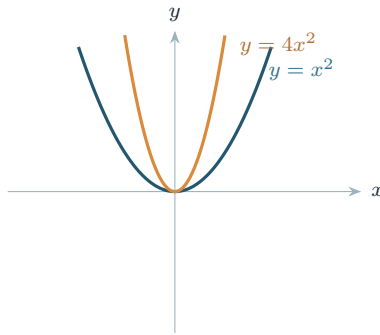
(a) $y = 3x^2$

(b) $y = -x^2 + 4$

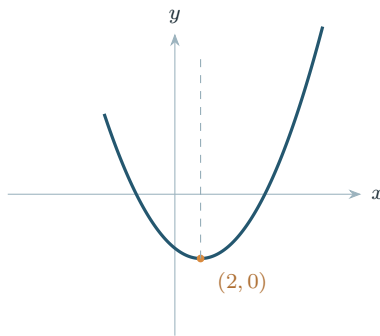
H3. State the turning point of $y = x^2 - 5$ and say whether it is a maximum or minimum.



H4. Compare $y = x^2$ and $y = 4x^2$. Which is narrower, and why?



H5. State the turning point of $y = (x - 2)^2$ and the equation of its axis of symmetry.



7B Sketching parabolas using transformations

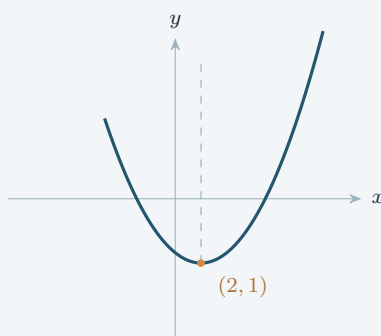
CORE

Textbook worked examples (cross-reference in your textbook)

- **Example 3** — Sketching with transformations (p. 597)
- **Example 4** — Using turning point form (p. 598)
- **Example 5** — Finding a rule from a simple graph (p. 598)

Question sets (practise from your textbook)

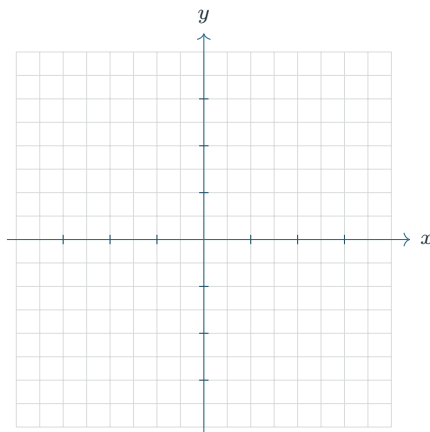
- **Building Understanding:** Q1–3 (p. 596)
- **Fluency:** Q1–4 (p. 600)
- **Problem-solving:** Q5–7 (p. 600–601)
- **Reasoning:** Q8–10 (p. 602)
- **Enrichment: Sketching with many transformations:** Q11 (p. 602)

Worked examples (study the method, then try the hurdles)**Sketching using turning point form**Sketch $y = (x - 2)^2 + 1$, labelling the turning point and y -intercept.

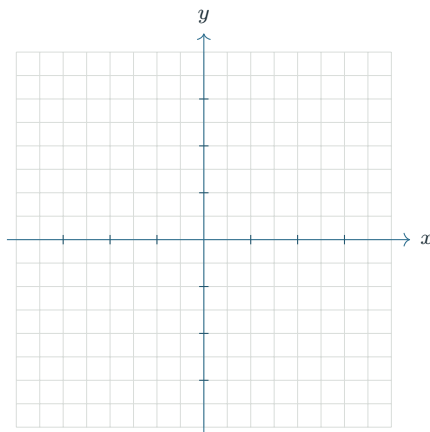
Solution. In the form $y = (x - h)^2 + k$ the turning point is $(h, k) = (2, 1)$, a minimum. For the y -intercept set $x = 0$: $y = (0 - 2)^2 + 1 = 5$. The parabola is upright with the same width as $y = x^2$.

Hurdle questions — Section 7B (show all working)

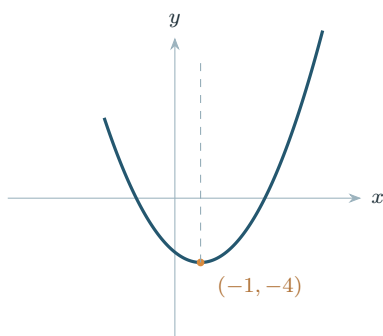
H1. Sketch $y = x^2 + 3$ on the grid, labelling the turning point and y -intercept.



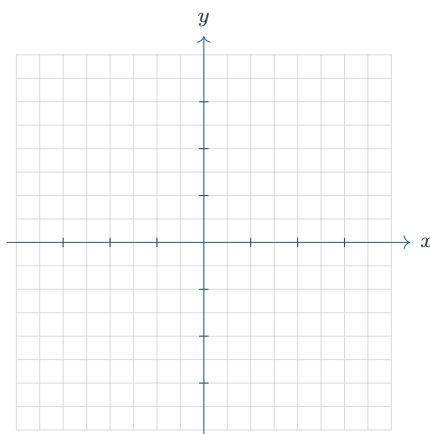
H2. Sketch $y = -x^2$ on the grid, labelling the turning point.



H3. State the turning point of $y = (x + 1)^2 - 4$, then state whether it is a maximum or minimum.



H4. Sketch $y = (x - 2)^2$ on the grid, labelling the turning point and y -intercept.



H5. A parabola has turning point $(3, -2)$ and is upright with the same shape as $y = x^2$. Write its rule in turning point form.

7C Sketching parabolas using factorisation

CORE

Textbook worked examples (cross-reference in your textbook)

- **Example 6** — Using the x -intercepts to find the turning point (p. 604)
- **Example 7** — Sketching a perfect square (p. 606)
- **Example 8** — Finding a turning point from a graph (p. 607)

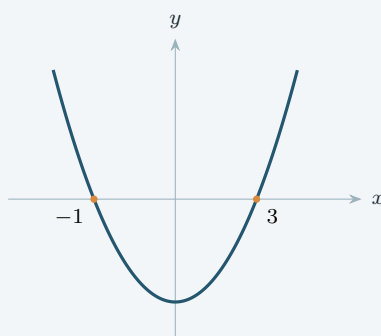
Question sets (practise from your textbook)

- **Building Understanding:** Q1–3 (p. 604)
- **Fluency:** Q1–5 (p. 608)
- **Problem-solving:** Q6–10 (p. 609)
- **Reasoning:** Q11–14 (p. 609)
- **Enrichment: More rules from graphs:** Q15 (p. 610)

Worked examples (study the method, then try the hurdles)

Sketching using factorisation

Sketch $y = x^2 - 2x - 3$, labelling the intercepts and turning point.



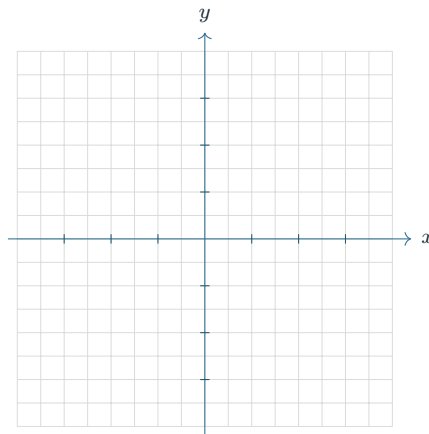
Solution. Factorise: $y = (x - 3)(x + 1)$, so the x -intercepts are $x = 3$ and $x = -1$. The axis of symmetry is midway between them: $x = \frac{3 + (-1)}{2} = 1$. Substituting $x = 1$: $y = 1 - 2 - 3 = -4$, so the turning point is $(1, -4)$. The y -intercept is -3 .

Hurdle questions — Section 7C (show all working)

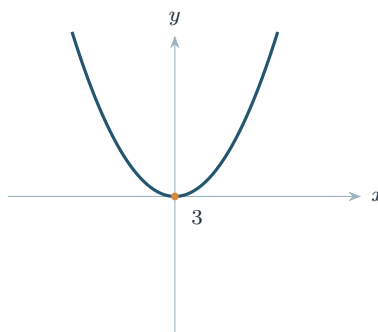
H1. Find the x -intercepts of $y = x^2 - 5x + 6$ by factorising.

H2. For $y = x^2 - 4x$, find the x -intercepts and hence the turning point.

H3. Sketch $y = (x - 1)(x - 5)$, labelling both x -intercepts, the y -intercept and the turning point.



H4. Sketch $y = x^2 - 6x + 9$, a perfect square. Label the single x -intercept and the turning point.



H5. Find the y -intercept and x -intercepts of $y = x^2 + 2x - 8$.

7D Sketching parabolas by completing the square

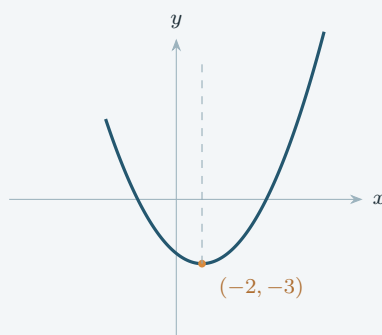
CORE

Textbook worked examples (cross-reference in your textbook)

- **Example 9** — Finding key features from turning point form (p. 612)
- **Example 10** — Sketching by completing the square (p. 613)

Question sets (practise from your textbook)

- **Building Understanding:** Q1–3 (p. 612)
- **Fluency:** Q1–6 (p. 615)
- **Problem-solving:** Q7–9 (p. 616)
- **Reasoning:** Q10–13 (p. 616)
- **Enrichment: Finding rules using the turning point:** Q14 (p. 617)

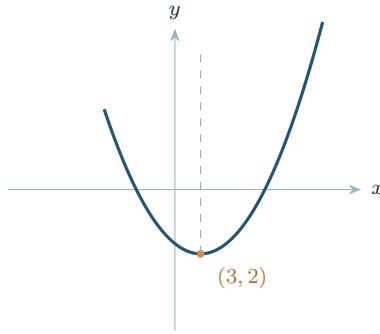
Worked examples (study the method, then try the hurdles)**Sketching by completing the square**Write $y = x^2 + 4x + 1$ in turning point form and state the turning point.**Solution.** Half of 4 is 2, and $2^2 = 4$:

$$\begin{aligned}y &= x^2 + 4x + 1 \\&= (x + 2)^2 - 4 + 1 \\&= (x + 2)^2 - 3.\end{aligned}$$

The turning point is $(-2, -3)$, a minimum; the y -intercept is 1.

Hurdle questions — Section 7D (show all working)

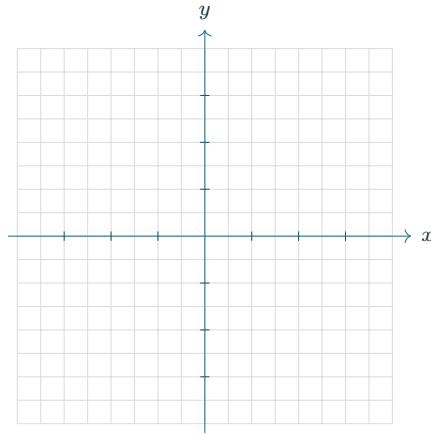
H1. State the turning point and y -intercept of $y = (x - 3)^2 + 2$.



H2. Write $y = x^2 + 4x + 1$ in turning point form by completing the square.

H3. Write $y = x^2 - 6x + 11$ in turning point form and state the turning point.

H4. Sketch $y = x^2 + 2x - 3$ by completing the square, labelling the turning point and y -intercept.



H5. State the minimum value of $y = x^2 - 10x + 30$ and the x -value where it occurs.

7E Sketching parabolas using the quadratic formula

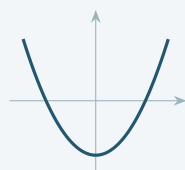
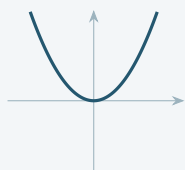
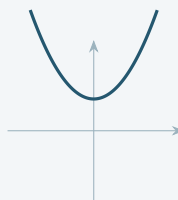
CORE

Textbook worked examples (cross-reference in your textbook)

- **Example 11** — Using the discriminant and $x = -\frac{b}{2a}$ (p. 620)
- **Example 12** — Sketching graphs using the quadratic formula (p. 621)

Question sets (practise from your textbook)

- **Building Understanding:** Q1–3 (p. 619)
- **Fluency:** Q1–4 (p. 622)
- **Problem-solving:** Q5–7 (p. 622)
- **Reasoning:** Q8–10 (p. 623)
- **Enrichment: Some proof:** Q11–12 (p. 623)

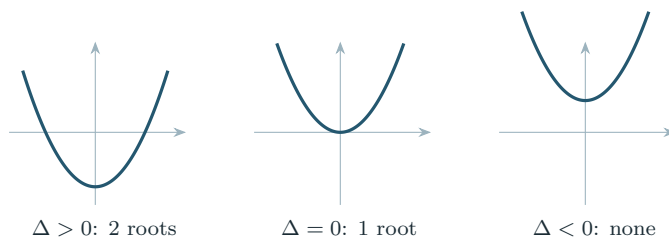
Worked examples (study the method, then try the hurdles)**Using the discriminant and the formula**For $y = x^2 - 4x + 2$, state the number of x -intercepts, then find them in surd form. $\Delta > 0$: 2 roots $\Delta = 0$: 1 root $\Delta < 0$: none**Solution.** $\Delta = (-4)^2 - 4(1)(2) = 16 - 8 = 8 > 0$, so there are **two** x -intercepts.

$$x = \frac{4 \pm \sqrt{8}}{2} = \frac{4 \pm 2\sqrt{2}}{2} = 2 \pm \sqrt{2}.$$

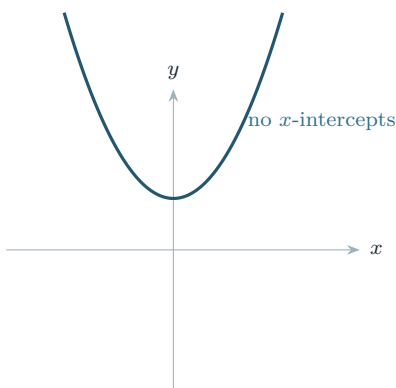
The axis of symmetry is $x = -\frac{b}{2a} = 2$.

Hurdle questions — Section 7E (show all working)

H1. For $y = x^2 - 4x + 1$, use the discriminant to state how many x -intercepts the parabola has.



H2. For $y = x^2 + 2x + 5$, use the discriminant to state how many x -intercepts the parabola has.



H3. Find the axis of symmetry of $y = x^2 - 6x + 4$ using $x = -\frac{b}{2a}$, then find the turning point.

H4. Find the x -intercepts of $y = x^2 - 2x - 2$ in exact (surd) form using the quadratic formula.

H5. Find the x -intercepts of $y = 2x^2 + 3x - 1$, correct to two decimal places.

7F Applications of parabolas

CORE

Textbook worked examples (cross-reference in your textbook)

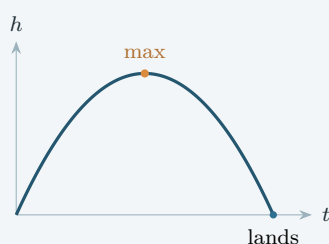
- **Example 13** — Applying parabolas given the rule (p. 626)
- **Example 14** — Applying parabolas by formulating a rule (p. 627)

Question sets (practise from your textbook)

- **Building Understanding:** Q1–2 (p. 626)
- **Fluency:** Q1–5 (p. 628–629)
- **Problem-solving:** Q6–8 (p. 630)
- **Reasoning:** Q9–12 (p. 630)
- **Enrichment:** The highway and the river and the lobbed ball: Q13–14 (p. 631)

Worked examples (study the method, then try the hurdles)**Applying a parabola**

A ball's height after t seconds is $h = -5t^2 + 30t$ metres. Find when it lands and its maximum height.



Solution. It lands when $h = 0$:

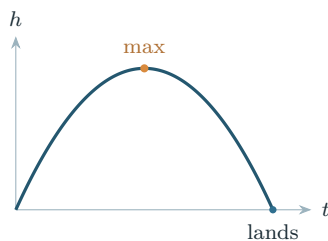
$$-5t(t - 6) = 0 \Rightarrow t = 0 \text{ or } t = 6,$$

so it lands after 6 seconds. By symmetry the maximum is at $t = 3$:

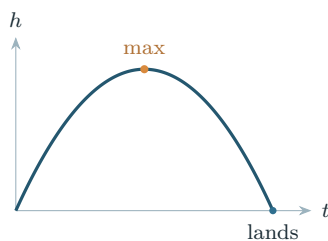
$$h = -5(9) + 30(3) = -45 + 90 = 45 \text{ m.}$$

Hurdle questions — Section 7F (show all working)

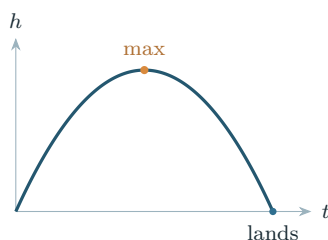
- H1.** A ball's height is $h = -5t^2 + 20t$ metres after t seconds. Find the times when the ball is at ground level.



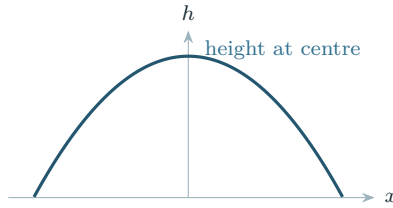
- H2.** For the ball in Question 1, find the time at which it reaches its maximum height.



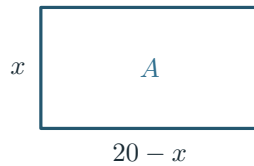
- H3.** For the ball in Question 1, find the maximum height reached.



- H4.** A parabolic arch has height $h = -\frac{1}{4}x^2 + 4$ metres, where x is the horizontal distance from the centre. Find the height at the centre and the width of the arch at ground level.



- H5.** A farmer has 40 m of fencing for a rectangular pen. If the width is x , show that the area is $A = x(20 - x)$, then find the x that gives the maximum area.



7G Intersection of lines and parabolas

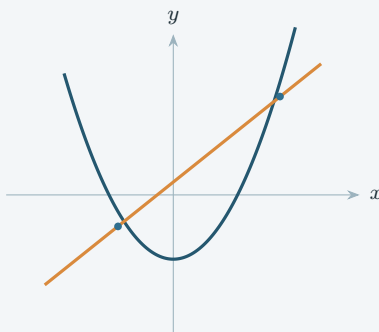
10A

Textbook worked examples (cross-reference in your textbook)

- **Example 15** — Intersection of a parabola and a horizontal line (p. 634)
- **Example 16** — Simultaneous equations: a line and a parabola (p. 635)
- **Example 17** — Simultaneous equations with the quadratic formula (p. 636)
- **Example 18** — Determining the number of solutions (p. 637)

Question sets (practise from your textbook)

- **Building Understanding:** Q1–3 (p. 633)
- **Fluency:** Q1–5 (p. 638)
- **Problem-solving:** Q6–8 (p. 639)
- **Reasoning:** Q9–11 (p. 639)
- **Enrichment: Multiple tangents?:** Q12 (p. 639)

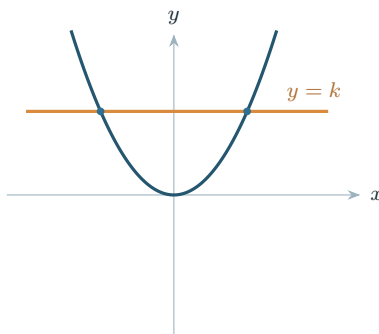
Worked examples (study the method, then try the hurdles)**Line meeting a parabola**Find the points where $y = x^2$ meets $y = x + 6$.**Solution.** Set the expressions equal and solve:

$$\begin{aligned}x^2 &= x + 6 \\x^2 - x - 6 &= 0 \\(x - 3)(x + 2) &= 0.\end{aligned}$$

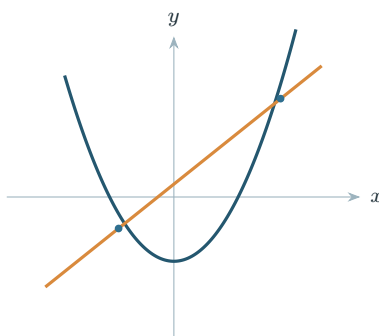
So $x = 3$ or $x = -2$, giving the points $(3, 9)$ and $(-2, 4)$.

Hurdle questions — Section 7G (show all working)

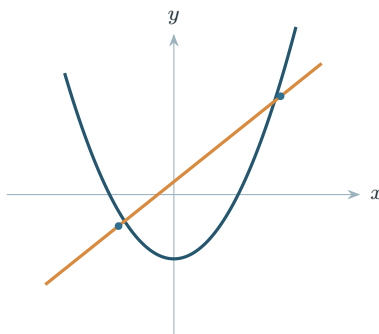
H1. Find the points of intersection of $y = x^2$ and the horizontal line $y = 9$.



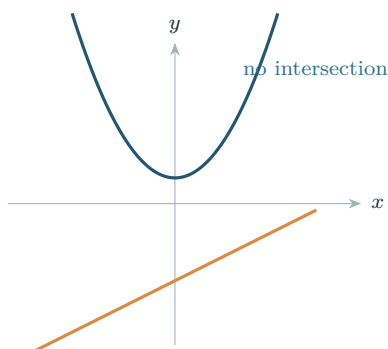
H2. Solve simultaneously to find where $y = x^2$ meets $y = x + 2$.



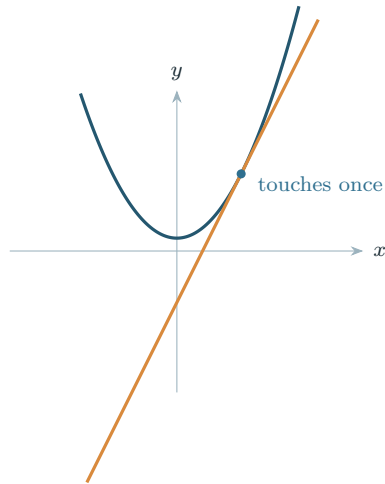
H3. Solve simultaneously to find where $y = x^2 - 3$ meets $y = 2x$.



H4. Use the discriminant to determine the number of points of intersection of $y = x^2$ and $y = x - 4$.



H5. Show that $y = x^2 + 1$ and $y = 2x$ touch at exactly one point, and find that point.



7H Rates of change

10A

Textbook worked examples (cross-reference in your textbook)

- **Example 19** — Graphing height vs time (p. 644)
- **Example 20** — Working with distance–time graphs (p. 645)

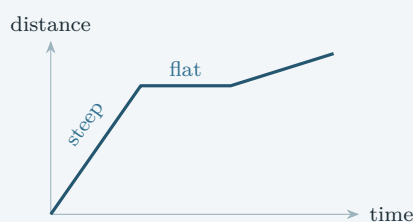
Question sets (practise from your textbook)

- **Building Understanding:** Q1–3 (p. 644)
- **Fluency:** Q1–6 (p. 647–648)
- **Problem-solving:** Q7–10 (p. 649)
- **Reasoning:** Q11–13 (p. 650)

Worked examples (study the method, then try the hurdles)

Reading a distance–time graph

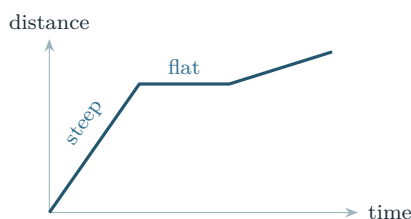
A distance–time graph rises steeply, then flattens to horizontal, then rises gently. Describe the motion.



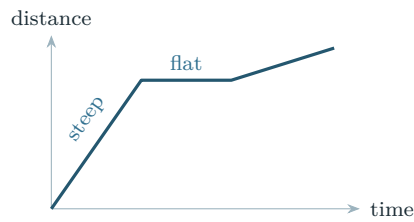
Solution. A *steep* section means a large distance in a short time, so the object is moving **fast**. A *horizontal* section means the distance is not changing, so the object is **stationary**. A *gentle* slope means it is moving **slowly**.

Hurdle questions — Section 7H (show all working)

H1. On a distance–time graph, state what a *steeper* section of the graph tells you about the speed.

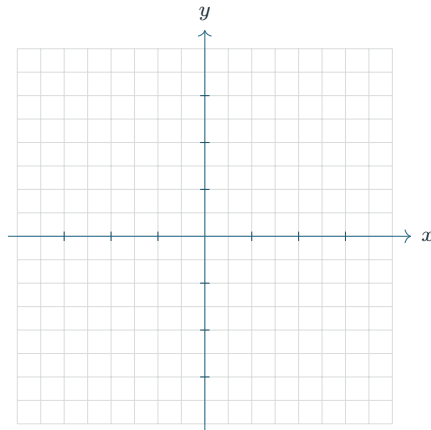


H2. A distance–time graph is a horizontal line for 3 seconds. Describe the motion during that time.

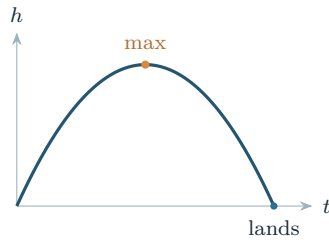


H3. A car travels 150 km in 2 hours. Find its average speed in km/h.

H4. Sketch a height–time graph for a ball thrown straight up and caught again, labelling the maximum height.



- H5.** For the height–time graph you sketched, state where the rate of change is zero and explain what is happening at that moment.



7I Average and instantaneous rates of change

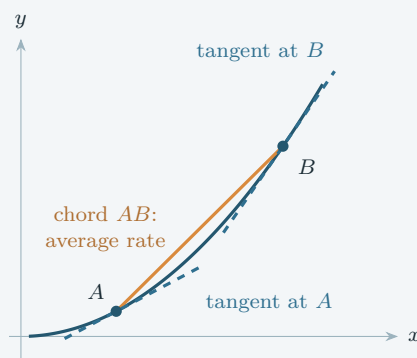
10A

Textbook worked examples (cross-reference in your textbook)

- **Example 21** — Finding an average rate of change (p. 654)
- **Example 22** — Approximating an instantaneous rate of change (p. 654)

Question sets (practise from your textbook)

- **Building Understanding:** Q1–3 (p. 653)
- **Fluency:** Q1–7 (p. 655–656)
- **Problem-solving:** Q8–11 (p. 657)
- **Reasoning:** Q12–14 (p. 658)
- **Enrichment: When an approximate rate is exact:** Q15 (p. 658)

Worked examples (study the method, then try the hurdles)**Average rate of change**For $y = x^2$, find the average rate of change between $x = 1$ and $x = 4$.

Solution. The average rate of change is the gradient of the chord joining the two points (1, 1) and (4, 16):

$$\frac{\text{change in } y}{\text{change in } x} = \frac{16 - 1}{4 - 1} = \frac{15}{3} = 5.$$

Hurdle questions — Section 7I (show all working)

H1. Explain the difference between an *average* rate of change and an *instantaneous* rate of change.

H2. For $y = x^2$, find the average rate of change between $x = 1$ and $x = 3$.

H3. For $y = x^2$, find the average rate of change between $x = 2$ and $x = 2.1$.

H4. On a curve, state which line gives the *average* rate of change and which gives the *instantaneous* rate of change: the chord or the tangent?

H5. A distance–time curve passes through $(2, 8)$ and $(5, 29)$. Find the average speed over this interval.

End of the quadratics hurdle booklet — check your working against the worked examples and ask in class if any hurdle question is unclear.